#Many Daughters
Impact of Having Daughters on Attitudes, Preferences, and Behaviors
Research Team ID:

Pre-Analysis Plan (PAP)

1. Study Information: Research Question and Hypotheses

This study investigates the causal impact of having daughters on gender-related outcomes using SOEP data. The following four hypotheses will be tested:

H1. *Having daughters leads to stronger support for gender equality (as measured by attitudes).*
That is, denoting X as a measure of support for gender equality,
(X for respondents with daughters) > (X for respondents without daughters).

H2. *Having daughters leads to more left-leaning political preferences.*
That is, denoting Y as a measure of left-leaning political preferences,
(Y for respondents with daughters) > (Y for respondents without daughters).

H3. *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).*
That is, denoting Z as a measure of gender equality in the household,
(Z for respondents with daughters) > (Z for respondents without daughters).

H4. *Having daughters leads to a larger increase in support of gender equality for men than for women.*
That is, denoting X as a measure of support for gender equality,
(X for male respondents with daughters − X for male respondents without daughters) >
(X for female respondents with daughters − X for female respondents without daughters).

2. Data and Data Access

The analysis is based on secondary data from the German Socio-Economic Panel (SOEP), a longitudinal household survey conducted annually by the German Institute for Economic Research (DIW Berlin). The SOEP collects comprehensive information on individuals and households living in Germany, including demographics, income, employment, education, and values. SOEP-Core data are currently available for the years 1984 to 2023 (v40, DOI: 10.5684/soep.core.v40eu). However, the proposed analyses are based on SOEP-Core v37, including observation years for 1984 to 2021, to exclude information surveyed during the COVID-19 pandemic.

Data access is provided by the DIW Berlin in accordance with the ManyDaughters Project, for which a 5% subsample of the SOEP data is made available under a data use agreement.

3. Variables

Outcome/Dependent Variables:

H1. & H4. Support for gender equality

To measure support for gender equality, we construct a standardized mean index based on four attitudinal items from survey years 2012 (pl-dataset; only in FiD subsamples L1, L2, and L3) and 2018 (pl-dataset; all subsamples excluding migration samples M3, M4, and M5):

1) "A child under the age of 6 suffers if their mother works." [plh0298_v1 in 2012 and plh0298_v2 in 2018]
2) "A person who is living with their partner for the long term should get married" [plh0300_v1 in 2012 and plh0300_v2 in 2018]
3) "A child under the age of three suffers if his or her mother works." [plh0302_v1 in 2012 and plh0302_v2 in 2018]
4) "It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally." [plh0308_v1 in 2012 and plh0308_v2 in 2018]

These items are part of a battery introduced with the following prompt:

> *"I will read you a series of statements. To what degree do you personally agree with each statement?" Please answer according to the following scale. 1 means disagree completely, and 7 means agree completely. You can use the in-between ratings to tailor your response.*

The index is calculated as the mean of the four standardized components, with the first three items being reverse-coded so that higher values indicate more egalitarian views, thus, stronger support for gender equality.

H2. Left-leaning political preferences

Political preferences are measured using the left–right self-placement question [plh0004] (syears 2005, 2009, 2014, and 2019; pl-dataset; excluding migration samples M3, M4, and M5), asking:

> *"In politics, people often talk about 'left' and 'right' when it comes to characterizing different political attitudes. If you think about your own political views: Where would you place yours?"*

Responses are given on an 11-point scale ranging from 0 (far left) to 10 (far right). For this study, we reverse-code the scale so that higher values indicate more left-leaning political preferences.

H3. Gender equality in the household

We operationalize gender equality in the household using a gender equality index based on the relative contribution of each partner to unpaid housework. This measure is derived from

harmonized time-use data on respondents' absolute hours typically spent on housework during a weekday (syears 1984 to 2020; pl-dataset; excluding: M1 (2013), M3 (2016, only partially collected for this sample in 2018), M4 (2016, only partially collected for this sample in 2018), M5 (2018), M6 (only partially collected)). Specifically, respondents are asked:

> *"What is a typical day like for you? How many hours do you spend on the following activities on a typical weekday?"*

While respondents provide responses for several tasks, we focus on the number of full hours for housework (laundry, cooking, cleaning) [pli0043_h], which is available for respondents prior to and after first childbirth. For each partner, we top-code the variable at the 99th percentile to deal with outliers.

We compute the gender equality index as follows:

$$Gender\ Equality\ Index = 1 - 2 * \left| \frac{Women's\ unpaid\ hours}{Women's\ unpaid\ hours + Her\ male\ partner's\ unpaid\ hours} - 0.5 \right|$$

This index ranges from 0 (one partner does all the housework) to 1 (perfectly equal division), with 0.5 indicating a moderate imbalance in contributions. Values closer to 1 reflect more equal sharing between partners, while values closer to 0 indicate a highly unequal distribution—regardless of whether the woman or the man does the majority of the work.This index ranges from 0 (one partner does all the housework) to 1 (perfectly equal division), with 0.5 indicating moderate inequality in either direction.

Explanatory/Independent Variables:

H1. & H4. Treatment indicator on whether respondent has a daughter

To test Hypotheses 1 and 4, we construct a binary indicator for whether a respondent has at least one biological daughter at the time of measurement of gender attitudes (i.e. in 2012 and/or 2018). This variable reflects exposure to having a daughter at the time attitudes were surveyed, which is central to our identification strategy. The indicator on having at least one biological daughter is derived from retrospective fertility histories available in the SOEP.

For each respondent and year, we define the variable as follows:

- has_daughter = 1 if the respondent has at least one daughter,
- has_daughter = 0 otherwise.

To test Hypothesis 4, which posits that having daughters affects men and women differently, we interact this treatment variable with a binary indicator for the respondent's gender. Gender is coded as a binary variable based on self-reported sex in the SOEP:

- gender = 1 for female respondents
- gender = 0 for male respondents

This interaction allows us to assess whether the effect of having daughters on gender attitudes differs by parent gender.

H2. & H3 Indicator of experiencing the birth of a first daughter

The key explanatory variable used to test Hypotheses 2 and 3 is a binary indicator for whether a previously daughter-less respondent has experienced the birth of a biological firstborn daughter at the time of observation. This variable is derived from retrospective fertility histories available in the SOEP.

For each respondent and year, we define the variable as follows:

- has_daughter = 1 if a biological firstborn daughter was born within the past year/years,
- has_daughter = 0 otherwise (i.e., no child or son is born).

This variable may vary over time as individuals transition from being childless or having only sons to becoming parents of a daughter. It is designed to capture exposure to having a daughter at a given point in time, which is the theoretical mechanism of interest in both hypotheses.

Covariates:

We include a set of control variables to account for potential confounding factors across our analyses. For **Hypotheses 1 and 4**, which rely on a between estimate, we include both time-varying and time-invariant covariates. In contrast, the analyses for **Hypotheses 2 and 3** are based on longitudinal data with individual fixed-effects models, which automatically account for all time-constant characteristics; therefore, only time-varying covariates are included in those models.

The following table provides an overview of all covariates used, including their justification, operationalization, and source in the SOEP data.

Covariate	Included in Hypotheses Test H?	Justification	Operationalization	Used variable(s) and dataset(s)
Respondent has a son/sons	H1, H4	Controls for the gender composition of children, allowing us to disentangle daughter effects from general parenthood effects or son-specific influences.	Binary: 0= no son 1= at least 1 son	**kidsex**(x) from biobirth
Respondent experiences birth of a son	H2, H3	Captures time-varying effects of son births to separate changes in attitudes or behaviors due to the birth of a son from those without births or due to the birth of a daughter.	Binary: 0= no son born 1= son born	**kidsex**(x) from biobirth
Age of first-born daughter	H1, H4	Controls for the time since treatment (birth of daughter), allowing for differentiation between immediate and long-term effects. Also accounts for possible age-related changes in parenting or social norms over time.	age in years, decimals (missing birth and interview months are replaced with "6, i.e. June")	**kidgeb**(x) from biobirth + **syear** and pgmonth from pgen
Age of youngest child in household	H2, H3	Captures the age and developmental stage of the child most likely to require active caregiving and labor supply, thus influencing division of labor and possibly attitudes; used to control for age-of-child effects as well as exposure to treatment (birth of a daughter).	Categorical: 0 no child 1=under three 2=three to under ten, 3=ten+ (missing birth and interview months are replaced with "6, i.e. June")	**kidgeb**(x) and kidmon(x) from biobirth + **syear** and **pgmonth** from pgen
Respondent's age and age squared	H1, H4, H2, H3,	Controls for life-cycle effects, as all outcomes may vary systematically with age. The squared term captures non-linear patterns (e.g., mid-life attitudinal shifts).	Years and years squared to capture also non-linear maturation effects. Missing birth years are replaced with information from previous waves. Missing birth and interview months are replaced with "6", i.e. June	**gebjahr** in ppathl + **Syear and pgmonth from pgen)**
Respondent's gender	H1	Respondents' gender is a major predictor of gender attitudes.	Binary: 0 = male 1 = female	**sex** from ppathl
Respondent's migration background	H1, H4	Accounts for differences in socialization, gender norms, or fertility preferences linked to migration history. May influence both likelihood of having daughters and outcomes.	Binary: 0 = none, 1 = has direct / indirect migration background	**migback** from ppathl

Respondent's siblings and their gender	H1, H4	Having siblings, particularly of a certain gender, may shape gender attitudes and preferences through early-life experiences and role models.	Binary: 0= no sibling 1= only male 2= only female 3= mixed	**nums** and **numb** from bioparen
Respondent's education	H1, H4	Educational attainment is a key predictor of more egalitarian gender attitudes and may confound the relationship between parenting daughters and support for gender equality.	Categorical: 0= low (ISCED=0, 2) 1= medium (ISCED=3,4) 2= high (ISCED=5,6)	**pgisced97** from pgen
Respondent's marital status	H1, H4, H2, H3,	Marital status can influence time use, household dynamics, and openness to gender norms. It may also interact with childbearing and gender ideology.	Binary: 0= other 1= married	**pgfamstd** and **pgpartz** in pgen
Respondent's cohabitation status	H1, H2, H4	Cohabiting with a partner can influence time use and household dynamics	Binary: 0= other 1= cohabiting	**pgpartz** in pgen
Respondent's employment status	H1, H4, H2, H3,	Employment may affect or reflect gender attitudes (H1, H4), ideology (H2), and capacity for unpaid labor (H3).	Categorical: 0= full-time (ref) 1= part-time 2= unemployed 3= other	**pgemplst** from pgen
Household income	H1, H4, H2, H3,	Income shapes opportunities and constraints in outsourcing unpaid labor and might correlate with preferences and values or structural possibilities to act on them.	Monthly net income, inflation adjusted (base year 2015), top-coded at the 99th percentile, and log-transformed and	**hghinc** from hgen, CPI from the Federal Statistical Office
Household's region (East/West)	H1, H4, H2, H3,	Reflects historically rooted differences in gender ideology, labor market participation, and family policy exposure between East and West Germany, which may confound relationships in H1 and H4.	Binary: 0= West Germany 1= East Germany	**hgnuts1**_ew from hgen
Household living in urban area	H1, H4, H2, H3,	Urban areas are often associated with more egalitarian gender norms and greater institutional support for dual-earner/female employment models.	Binary: 0= Rural 1= Urban	**regtyp** from hbrutto
Mode of interview	H1, H4, H2, H3,	The mode may influence responses due to social desirability or interviewer presence, with potential implications for how outcomes and covariates are reported.	Categorical: 0 = CAPI / CAWI 1 = with Interviewer	**pgmode** in pgen

Gender of interviewer	H1, H4, H2, H3,	Respondents may respond differently to male vs. female interviewers, particularly on gender- and ideology-related questions.	Binary: 0 = male 1 = female	**gender** in interviewer
Family policy context at interview	H2, H3	Time-fixed contextual controls capture broader structural conditions during the time of childbirth, such as policy reforms or cultural shifts, that may systematically influence both parenting and attitudes.	Categorical: 0= 2007/max 1= 2000/2006 2= 1984/1999	Based on **syear**
Survey year	H1, H4	Using a pooled cross-sectional design, it is relevant to account for the survey year	Binary: 0= 2018 1= 2012	**syear**

4. Sample and Inclusion/Exclusion Criteria

The analytical samples vary by hypothesis, depending on the outcome variables, measurement waves, and model design. However, a consistent set of baseline restrictions applies to all analyses. Based on all available survey waves (1984 to 2020) we restrict the sample to:

1) Respondents from all random GSOEP subsamples and associated refreshment subsamples, thus, excluding oversamples (variable "psample" from the ppathl dataset). Specifically, we exclude subsamples B, D, G, I, L2, L3, M1, M2, M3, M4, M5, P, Q, M6, M7, and M8, referring, for instance, to the high-income households, refugees, migrants or LGB individuals. We excluded these oversamples as they were designed to overrepresent specific groups, which could introduce distortions when analyzing broader patterns. In addition, not all examined outcome are sample for specific subsamples, such as the migration samples M1, M3, M4, M5, and M6). While these subsamples are valuable for subgroup-specific research, our focus is on trends across the general GSOEP sample. Moreover, key variables—particularly attitudes, time-use, and interviewer characteristics—are missing or inconsistently collected in these samples.
2) Survey years in which individuals live in private households (if variable "pop" from the ppathl dataset is smaller than 3) and no shared accommodation (hgowner not equal 5)
3) Only respondents' survey years in which respondents were successfully interviewed, ensuring that all variables are directly observed and not carried forward or imputed (if variable "netto" from the ppathl dataset is smaller 20).
4) Respondents' observations until the survey year 2020 to restrict our analyses to pre-COVID-19 pandemic times.
5) Individuals in different-sex partnerships (based on respondents' and partners' sex), retaining observations with no partner in the household.
6) Respondents aged 18 and over, to focus on individuals likely to enter parenthood (generating an age variable using the variable "gebjahr" from the ppathl dataset and the variable "syear").
7) Respondents are ever observed as being partnered during their participation in the SOEP. We follow these respondents as soon as they enter their first partnership during their SOEP participation. Specifically, we identify the first year each respondent is observed in a cohabiting or married relationship and exclude all person-years prior to this initial partnership. Respondents who are never observed with a partner or who enter the panel as singles are dropped.

Further Outcome/Dependent Variables-specific sample restrictions:

Below, we outline the specific inclusion and exclusion criteria applied for each hypothesis.

H1. & H4. Support for gender equality (attitudes):

These hypotheses are tested using a pooled-cross-sectional design, relying on the survey years 2012 and 2018.

1) Although all available survey years are used for the generation of variables, the analytical sample is restricted to waves 2012 and 2018, as outcome variables were only measured in the respective two waves. However, in these two years, gender attitudes were not measured across all SOEP samples. Therefore, our sample is automatically restricted to L samples for the year 2012 but includes all core samples for the year 2018.
2) The sample includes only respondents with non-missing responses to all analytical variables.

H2. Left-leaning political preferences

This hypothesis is evaluated using a linear regression model with individual fixed-effects, leveraging within-individual changes over time. The outcome measure is collected in the survey years 2005, 2009, 2014, and 2019. As such, our sample consists of respondents who (I) transition to parenthood of a (first) daughter between 2005 to 2020, or (II) who stay without a daughter during this time (i.e. remain either childless or experience the birth of a son/sons over time). The sample is constructed according to the following steps:

1) Because the outcome variable—political ideology self-placement on a left–right scale—is available in the 2005, 2009, 2014, and 2019 survey waves, the sample is restricted to those waves. However, all available SOEP waves were used for the generation of other analytical variables.
2) Observations with missing values on any of the analytical variables are excluded.
3) We restrict the sample to childless individuals or individuals with son(s) at first observation.
4) We exclude respondents aged 56 and over, to focus on individuals likely to enter parenthood.
5) The sample is restricted to respondents with valid responses in at least two of the four waves, allowing for within-person change to be estimated. Respondents who experience the birth of a daughter, have to be observed at least once before and once after the birth.

H3. Gender equality in the household

This hypothesis is evaluated using a panel fixed-effects design, leveraging within-individual changes over time. The outcome measure is collected annually. As such, our sample consists of respondents who (I) transition to parenthood of a daughter/daughters between the start of their first observation and 2020, or (II) who stay without a daughter/daughters during this time (i.e. are either childless or have a son/sons over time). The sample is constructed according to the following steps:

1) Observations with missing values on any of the analytical variables are excluded.

2) Respondents without valid partner information on housework are excluded in order to construct the measure on gender equality in the household. Hence, we focus only on observations in which respondents are partnered.
3) We restrict the sample to childless individuals or individuals with son(s) at first observation.
4) We exclude respondents aged 56 and over to focus on individuals likely to enter parenthood.
5) The sample is restricted to respondents with valid responses in at least two waves, allowing for within-person change to be estimated. Respondents who experience the birth of a daughter have to be observed at least once before and once after the birth.

5. Analysis

Statistical models

We will conduct four sets of analyses to examine the four hypotheses. All analyses will be performed using Stata (version 18). Our analytical strategy varies by hypothesis and is tailored to the structure of the outcome variables and the identification challenges each hypothesis presents. For each hypothesis and analyses, we state upfront which model and variables, i.e. outcomes (dependent variables), predictors (independent variables) and controls (covariate variables) are applied and how standard errors are estimated. Each set of models is described in more detail below.

H1. & H4. Support for gender equality (attitudes):

To test Hypotheses 1 and 4, we estimate pooled ordinary least squares (POLS) regression models. We rely on OLS rather than an instrumental variable approach or other methods because the sex of a child in Germany is essentially random and not subject to parental selection based on gender attitudes. As such, treatment assignment (i.e., having a daughter as firstborn child) can be considered plausibly exogenous in this context. Conditioning on observable characteristics further strengthens the credibility of our identification strategy.

For **Hypothesis 1**, the model is estimated using pooled ordinary least squares (POLS):

$$GenderAttitudes_{it} = \alpha + \beta\, TreatDaugher_{it} + X_{it}\gamma + \varepsilon_{it}$$

Where:

- $GenderAttitudes_i$: Standardized index of gender attitudes
- α: Intercept
- $TreatDaugher_i$ = 1 if respondent has daughter, $TreatDaugher_i$ = 0 otherwise (i.e. childless or son(s) only)
- X_i: Vector of covariates (e.g., individual-, household- and context-level controls —see covariate section)
- γ: Vector of coefficients corresponding to the covariates

- ε_i: Error term

We apply sampling weights and estimate robust standard errors, clustered at the individual level, to account for heteroskedasticity in the cross-sectional sample. Overall, the coefficient β estimates the average difference in gender attitudes between those exposed to raising a daughter and those not (yet) exposed at the time of measurement.

To test **Hypothesis 4**, we extend the POLS from Hypothesis 1 by including an interaction between the treatment indicator and respondent gender. This allows us to test whether the effect of having a daughter differs significantly between male and female respondents.

We estimate the following model using ordinary least squares (OLS):

$$GenderAttitudes_{it} = \alpha + \beta_1 TreatDaugher_{it} + \beta_2 Female_{it} + \beta_3 (TreatDaugher_{it}\ x\ Female_{it}) + X_{it}\gamma + \varepsilon_{it}$$

Where:

- $Female_i$ =1: If respondent is female, = 0 if male
- $TreatDaugher_i\ x\ Female_i$: Interaction term for testing heterogeneity
- All other aspects of the equation are defined above

The coefficient β_3 captures the difference in the effect of having a daughter between women and men. A significant positive β_3 would indicate that women shift more toward gender-equal attitudes than men do, conditional on covariates.

H2. Left-leaning political preferences

To test Hypothesis 2, we use a two-way fixed effects (TWFE) model that accounts for both individual-specific and time-specific unobserved heterogeneity. This approach allows us to estimate how becoming a parent of a first-born daughter changes an individual's political preferences, while adjusting for all time-invariant characteristics as well as selected time-varying factors and year-specific context effects (e.g., family policies). This is important because the outcome—political self-placement—is potentially influenced by broader temporal dynamics, such as public discourse, social movements, or political events that affect all respondents in a given year. The TWFE design helps isolate the effect of having a daughter from such confounding influences.

We estimate the following model:

$$LeftPref_{it} = \beta HasDaughter_{it} + X_{it}\gamma + \alpha_i + \zeta_t + \varepsilon_{it}$$

Where:

- $LeftPref_{it}$: Reverse-coded left–right political placement, for individual i in year t

- $HasDaughter_{it}$: Binary indicator equal to 1 if individual i has at least one biological daughter in year t, 0 otherwise (no childbirth or birth of a son)
- X_{it}: Vector of time-varying covariates
- γ: Vector of coefficients corresponding to the covariates
- α_i: Individual fixed effects (captures all unobserved time-invariant characteristics)
- ζ_t: Year fixed effects (captures period shocks and general trends)
- ε_{it}: Error term

Standard errors are clustered at the individual level to account for autocorrelation and heteroskedasticity across time within respondents. We do not apply weights.

A positive β would suggest that having a daughter shifts respondents' political orientation to the left, while a negative β would indicate a shift toward more right-leaning preferences.

H3. Gender quality in the household

Hypothesis 3 is tested using the same two-way fixed effects (TWFE) framework from H2, applied to a behavioral outcome measuring the division of unpaid household labor between partners. By differencing out individual- or couple-level fixed characteristics, the model addresses potential bias from unmeasured stable traits. Again, period fixed effects capture general social and economic trends affecting all households.

We estimate the following model:

$$FemaleShare_{it} = \beta HasDaughter_{it} + X_{it}\gamma + \alpha_i + \zeta_t + \varepsilon_{it}$$

Where:

- $FemaleShare_{it}$: The female partner's share of total unpaid labor (housework, childcare, caregiving) in couple i in year t; ranges from 0 (no unpaid work done by woman) to 1 (all unpaid work done by woman)
- $HasDaughter_{it}$: Binary indicator equal to 1 if individual i has at least one biological daughter in year t, 0 otherwise
- X_{it}: Vector of time-varying covariates
- γ: Vector of coefficients corresponding to the covariates
- α_i: Couple fixed effects (captures all unobserved time-invariant characteristics)
- ζ_t: Year fixed effects (captures period shocks and general trends)
- ε_{it}: Error term

We use clustered standard errors at the couple level, to account for serial correlation and heteroskedasticity. No survey weights are applied.

Lower values of the dependent variable indicate a less equal distribution of unpaid labor (i.e., a greater burden on the female partner). A negative β would suggest that having a daughter reduces the female share — that is, moves the household toward more gender-equal behavior.

6. Interpretation of Results and Inference Criteria

Each hypothesis will be tested using a regression model tailored to the outcome variable and analytical design, as detailed in the Statistical Analysis section. For each hypothesis, we will present one regression table containing:

- Estimated coefficients for the key independent variable(s)
- Robust or cluster-robust standard errors (depending on model structure)
- Corresponding test statistics (typically t-values)
- p-values for significance testing
- Additional model diagnostics (e.g., fixed effects, number of observations, R-squared)

To assess whether each hypothesis is supported, we will interpret the following:

1. Direction of the estimated effect: Does it align with the theoretical expectation?
2. Statistical significance (p-value): Is the estimate distinguishable from zero, using conventional thresholds (e.g., $p < 0.05$)?
3. Magnitude of the effect: Is the estimated coefficient substantively meaningful in the context of the outcome?
4. Precision: Do confidence intervals suggest a narrow and interpretable range of plausible values?

While p-values will be reported to assess statistical significance, we emphasize that they do not indicate the importance or size of an effect. Therefore, we interpret our results primarily through the magnitude, direction, and confidence intervals of the estimated coefficients.

Hypothesis-Specific Interpretation

- H1 (Gender attitudes): A positive coefficient on the treatment variable (first daughter born before 2018) supports the hypothesis that having a daughter leads to stronger support for gender equality.
- H2 (Political preferences): A positive effect on political self-placement indicates a shift toward more left-leaning views, consistent with the hypothesis.
- H3 (Household labor): A negative coefficient on the indicator for having a daughter would reflect a decline in the female share of unpaid work, signaling increased gender equality in the household.
- H4 (Gender heterogeneity in attitudes): The interaction term between treatment and respondent gender captures differential effects. A positive and significant interaction supports the hypothesis that men respond more strongly to having a daughter than women do.